

ENTERPRISE LOGISTICS ARCHITECTURE & TECHNICAL DOCUMENTATION

COURIER with NETPACK

A comprehensive, world-class standard logistics management system for International Air Cargo Hub/Agency routing, Nepal Domestic Highway express operations across 77 districts, automated tracking telemetry, and certified HAWB generation.



Figure 1.0: NETPACK International Air Cargo Flight Departure from Kathmandu Tribhuvan Gateway (KTM) connecting with Global Aviation Corridors

System Version	NETPACK v2.0 Enterprise (Release 2026)	Primary Stack	PHP 8.3+ · Laravel 12 · MariaDB/MySQL 8
International Scope	IATA HAWB · MAWB Airline Flights · Overseas Hubs (DXB, LHR, JFK, SYD)	Carrier Webhooks	FedEx · DHL · UPS · Royal Mail · Australia Post · DPD · Aramex

Domestic Coverage	7 Provinces · 77 Districts · Highway Linehauls · Ward Riders	Test Suite Integrity	111 Feature & Unit Tests (100% Passing · 771 Assertions)
Author & Maintainer	Engineering & Global Operations Team	Classification	Official System Architecture Reference Manual

Table of Contents

1. Executive Overview & Core Mission	Section 1
2. Technology Stack & Enterprise Architecture	Section 2
3. User Role Hierarchy & Operational Portal Matrix	Section 3
4. End-to-End Shipment Lifecycle & Master Flowchart	Section 4
5. International Air Cargo & Hub/Agency System (MAWB / HAWB)	Section 5
6. Global Last-Mile Carrier Sync & Inbound Webhooks Pipeline	Section 6
7. Nepal Domestic Highway Logistics (7 Provinces & 77 Districts)	Section 7
8. Booking & Scheduled Pickup Lifecycle Automation	Section 8
9. Universal Multi-Identifier Tracking & Privacy-Safe API	Section 9
10. Official House Air Waybill (HAWB) & Consignment Note Printing	Section 10
11. Database Schema & Key Entity Relationship Models	Section 11
12. Operational Runbook, Deployment Guards & Test Suite Verification	Section 12

1. Executive Overview & Core Mission

COURIER with NETPACK is a modern, unified logistics and courier management platform engineered to resolve the unique logistical challenges of Nepal while providing seamless, automated connectivity to global cargo corridors. The system operates on an advanced *Hub-and-Spoke* architecture spanning international freight consolidation, Master Air Waybill (MAWB) flights, overseas agency breakdown gateways, domestic highway linehauls across Nepal's 7 Provinces, and last-mile ward-level delivery riders with instant Cash on Delivery (COD) settlement.

Strategic Capabilities at a Glance:

- **Dual-Track Operations:** Seamlessly manages both International Air Freight (IATA HAWB/MAWB) and Domestic Express across all 77 districts of Nepal under one unified administrative umbrella.
- **Automated Telemetry Cascades:** Updating a Master Air Waybill (MAWB) flight immediately cascades departure, hub arrival, and customs clearance checkpoints to hundreds of bundled child consignments.
- **Global Carrier Telemetry Sync:** Automated integration with Tier-1 carriers (FedEx, DHL, UPS, Royal Mail, Australia Post, DPD, Aramex) via automated polling and inbound webhooks.
- **Printable Official HAWBs:** Built-in zero-charges customs-compliant House Air Waybill copies with QR code and 1-click A4 multi-part printouts accessible directly from public and administrative tracking portals.

2. Technology Stack & Enterprise Architecture

LAYER / COMPONENT	TECHNOLOGY	ARCHITECTURAL PURPOSE
Backend Runtime	PHP 8.3+ · Laravel 12.x	Enterprise MVC engine, Eloquent ORM, Service Layer, and event-driven architecture.
Database & Storage	MariaDB 10.4+ / MySQL 8.0 · S3 / Local	ACID transaction guarantees, spatial coordinates, atomic sequences, and encrypted document storage.
Frontend & UI	Blade, Tailwind CSS, Vanilla JS, Vite	High-performance responsive UI, dynamic status steppers, and mobile-friendly scan desks.
Geospatial & Radar	Leaflet.js · OpenStreetMap Tiles	Interactive international flight curves (KTM ? Hub ? Destination) and Nepal highway routes.
Barcode & Barcode/QR	Endroid QR Code · Milon Barcode	High-density dynamic QR codes resolving to tracking endpoints and Code128 barcodes on manifests.
PDF Generation	Barryvdh DomPDF · Headless Print Engine	Pixel-perfect multi-copy A4 HAWBs, manifests, POD consignment notes, and datasheets.
Async Queues & Crons	Laravel Queue Workers · Artisan Scheduler	Carrier telemetry polling, SMS/email milestone alerts, and SLA delivery reminders.

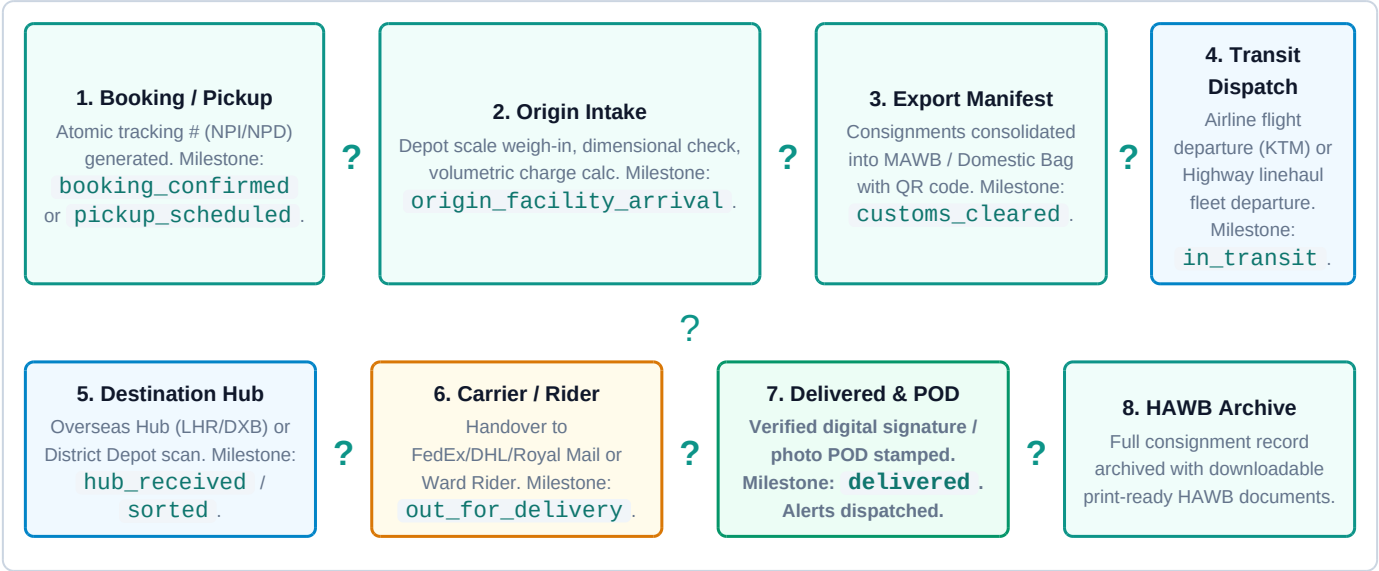
3. User Role Hierarchy & Operational Portal Matrix

The platform enforces strict role-based access control (RBAC) across 10 distinct user types to ensure complete operational data segregation and privacy compliance:

USER TYPE	PORTAL ROUTE	OPERATIONAL RESPONSIBILITIES & SCOPE
Super Admin	/admin/dashboard	Complete platform oversight, rate matrix configurations, godown handling charges, staff management, and system-wide audits.
International Admin	/international/dashboard	Air cargo rates, overseas hubs, agency formats, MAWB flight assignments, export manifests, and global carrier bindings.
Domestic Admin	/domestic/dashboard	District partner hubs, provincial routing linehails, vehicle manifests, scan desk audits, and delivery SLA enforcement.
Operations Staff	/domestic/dashboard	Package intake, scale weigh-in verification, barcode bag consolidation, and manifest loading.
Overseas Partner / Hub	/overseas/dashboard	Overseas gateway arrival notice processing (whole/partial), customs breakdown, and last-mile carrier handover.
Domestic Partner	/partner/dashboard	Regional district depot operations (e.g. Pokhara, Biratnagar, Chitwan), bag reception, and rider dispatch.
Delivery Rider	/rider/dashboard	Mobile runsheet, doorstep ward delivery, verified photo/signature Proof of Delivery (POD), and instant COD collection.
E-Commerce Seller	/seller/dashboard	Bulk order uploads, label printing, live dispatch status radar, and COD wallet payout settlements.
Business Client	/client/dashboard	Active shipment radar, bulk rate calculator, HAWB copies repository, and monthly account billing.
Public / Customer	/tracking/{number}	Universal tracking lookup, interactive flight/highway radar, milestone alert subscription, and HAWB printout.

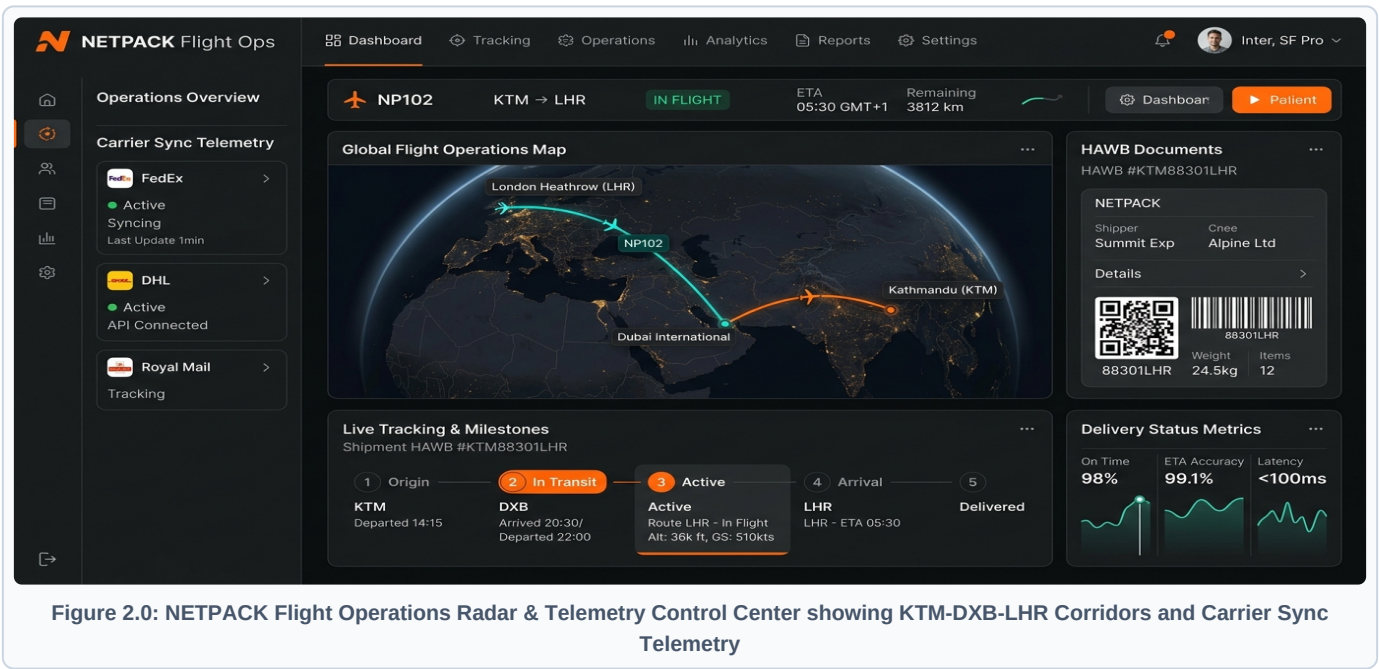
4. End-to-End Shipment Lifecycle & Master Flowchart

The following workflow chart outlines the automated milestone propagation across the entire operational lifespan of a consignment:



5. International Air Cargo & Hub/Agency System

Netpack's international cargo division manages air waybill processing from Kathmandu's Tribhuvan International Airport (KTM) to premier overseas distribution gateways.



Master Air Waybill (MAWB) Automated Cascades

The platform implements the `AutomatedTrackingService::cascadeMawbMilestone()` engine. When an international administrator updates a Master Air Waybill (e.g. Emirates flight EK-235 from KTM to Dubai DXB), the system automatically performs an atomic query across all bound child consignments:

Automated MAWB Event Progression:

- `in_transit` : Child shipments inherit `in_transit_airline` with airline name, flight number, origin KTM, and hub destination.
- `cleared` : Child shipments inherit `customs_cleared` at overseas gateway.
- `completed` : Child shipments inherit `hub_received` at overseas facility, preparing for local carrier transfer.

6. Global Last-Mile Carrier Sync & Inbound Webhooks Pipeline

For cross-border shipments, Netpack delegates final doorstep delivery to premier global carriers. The `CarrierTrackingSyncService` normalizes diverse carrier statuses into a standardized Netpack milestone schema:

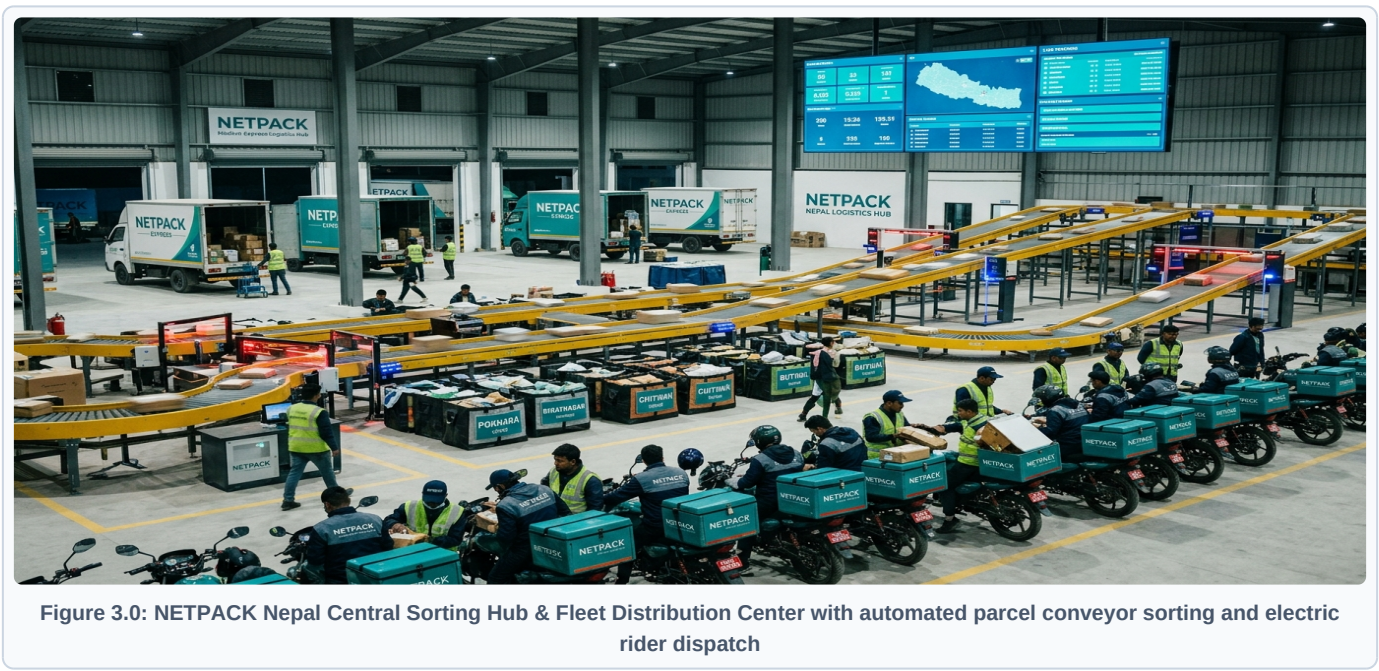
GLOBAL CARRIER	COVERAGE REGION	TELEMETRY PROTOCOL	STANDARDIZED EVENT MAPPING
FedEx	North America, Global	REST API & Webhook (JSON)	DELIVERED ? delivered , ON_WAY ? in_transit
DHL Express	Europe, Asia-Pacific, Middle East	Webhooks & Polling (XML/JSON)	delivered ? delivered , transit ? in_transit
Royal Mail	United Kingdom	Track & Trace API (JSON)	Delivered ? delivered , Collected ? picked_up
Australia Post	Australia, New Zealand	Shipping & Tracking API	Delivered ? delivered , In transit ? in_transit
DPD Group	Europe Continental	Webhook & REST Telemetry	Delivered ? delivered , Out for Delivery ? out_for_delivery
Aramex	GCC, Middle East, Africa	SOAP / REST Webhook	Delivered ? delivered , Forwarded ? in_transit

Inbound Webhook Endpoint: POST /api/webhooks/carrier-tracking/{carrier}

Payload includes carrier tracking number, status, location, timestamp, and signature confirmation. Updates the shipment status and instantly dispatches customer milestone notifications.

7. Nepal Domestic Highway Logistics (7 Provinces & 77 Districts)

The domestic network connects Kathmandu Central Sorting Hub with all regional distribution centers across Nepal via dedicated highway express linehauls and local partner depots.



Provincial Network Corridors

PROVINCE	PRIMARY REGIONAL SORTING HUB	SAMPLE DISTRICTS SERVED	LINEHAUL HIGHWAY TRANSIT
Province 1 (Koshi)	Biratnagar Regional Hub (BRT)	Morang, Sunsari, Jhapa, Ilam, Dhankuta	East-West Highway Express (14-18 hrs)
Province 2 (Madhesh)	Janakpur / Birgunj Central Hub	Dhanusha, Parsa, Bara, Sarlahi, Siraha	B.P. Highway / Terai Corridor (8-12 hrs)
Province 3 (Bagmati)	Kathmandu Central Sorting Hub (KTM)	Kathmandu, Lalitpur, Bhaktapur, Chitwan, Kaski	Prithvi Highway / Ring Road Express (1-4 hrs)
Province 4 (Gandaki)	Pokhara Regional Hub (PKR)	Kaski, Tanahun, Syangja, Baglung, Gorkha	Prithvi Highway Fleet (6-8 hrs)
Province 5 (Lumbini)	Butwal / Bhairahawa Gateway	Rupandehi, Kapilvastu, Dang, Palpa, Banke	Siddhartha & East-West Highway (9-12 hrs)
Province 6 (Karnali)	Surkhet Logistics Depot	Surkhet, Dailekh, Jumla, Salyan	Karnali Highway Linehaul (18-24 hrs)
Province 7 (Sudurpashchim)	Dhangadhi Regional Hub	Kailali, Kanchanpur, Doti, Dadeldhura	Mahakali & East-West Corridor (20-26 hrs)

Consolidated Manifest Bags & Scan Desk

Domestic consignments are consolidated into tamper-evident nylon bags tagged with unique QR codes. Hub operators process bags at the Scan Desk (/domestic/manifests/scan) with automated atomic cascades:

- **Arrival Scan:** Stamping a bag as arrived at Pokhara Hub automatically marks all 50+ enclosed shipments as `in_transit` at Pokhara Regional Sorting Hub.

- **Rider Dispatch:** Scanning an individual consignment assigns the local ward rider and stamps the milestone as `out_for_delivery`.
- **POD Capture:** The rider portal captures customer digital signature, recipient name, and delivery timestamp.

8. Booking & Scheduled Pickup Lifecycle Automation

The platform eliminates clerical delays by initializing tracking records the second an order or pickup request is created:

Automated Pickup Lifecycle:

- Request Created:** Model hook in `PickupRequest::booted()` allocates an atomic tracking number (e.g. `NPD-2026-000101-5`) and seeds event `pickup_scheduled`.
- Courier Assigned:** Dispatcher assigns rider; system stamps milestone `pickup_assigned` with courier name and contact.
- Physical Collection:** Rider scans parcel at customer doorstep; milestone automatically transitions to `shipment_picked_up`.
- Depot Weigh-In:** Package checked into Kathmandu Gateway; scales verify weight; tracking milestone advances to `origin_facility_arrival`.

9. Universal Multi-Identifier Tracking & Privacy-Safe API

Customers and logistics partners can locate any consignment using any known identifier through the universal search algorithm in `TrackingController::show()`:

IDENTIFIER TYPE	SAMPLE SYNTAX	RESOLUTION ENGINE
Netpack Tracking Number	<code>NPI-2026-000101-7</code> , <code>NPD-...</code>	Direct atomic lookup with hyphen-insensitive matching.
Regional HAWB Number	<code>UKNP-2026-001</code> , <code>USNP-2026-042</code>	Resolves child shipment under regional agency code.
Master Air Waybill (MAWB)	<code>176-98765432</code>	Resolves flight air manifest and active consignment cluster.
Global Carrier Number	<code>RM123456789GB</code> , <code>1Z99999999</code>	Resolves international parcel via last-mile carrier tracking number.
Pickup Request Number	<code>NPD-2026-000088-2</code>	Resolves pickup status and scheduling timeline.

Privacy-Safe Public API Endpoint

Endpoint: `GET /api/v1/track/{trackingNumber}`
Provides full event timestamps, location history, and estimated delivery dates while strictly redacting sensitive personal phone numbers, detailed street addresses, and monetary amounts.

10. Official House Air Waybill (HAWB) & Consignment Note Printing

Every tracked shipment provides immediate access to print-ready official documentation:

Available Printable Documents:

- International HAWB (A4 Multi-Part):** Route `/tracking/{number}/hawb` renders standard IATA multi-part copies (Consignee Copy, Customs/Airline Operations Copy, Carrier Copy) with zero charges (customs compliant), IATA liability disclaimers, and scannable tracking QR code.
- Domestic Waybill / Consignment Note:** Renders runsheet and POD carrier copies with Nepal highway route points and destination ward tags.
- Single-Page Quick Slip:** Route `/tracking/{number}/hawb/print` renders compact label-printer slips for rapid warehouse dispatch.

11. Database Schema & Key Entity Relationship Models

The database architecture is designed with strict foreign key constraints, atomic sequence generators, and JSON telemetry fields:

TABLE NAME	PRIMARY KEYS & REFERENCES	KEY COLUMNS & PURPOSE
shipments	id, customer_id, mawb_id, hub_id	Core international/standard consignment records. Contains tracking_number, hawb_number, last_mile_tracking_number, tracking_history (JSON), status.
domestic_shipments	id, client_id, partner_id	Domestic courier consignments across 77 districts. Contains receiver_ward, receiver_zone, weight, tracking_history, status.
m_a_w_b_s	id, hub_id	Master Air Waybill airline records. Contains mawb_number, airline_name, flight_number, origin_airport, destination_airport, status.
overseas_hubs	id	International gateway hubs (e.g. LHR, DXB, JFK). Contains hub_name, hub_code, country, is_active.
manifests & manifest_bags	id, partner_id, manifest_id	Inter-provincial linehaul manifests and nylon consolidation bags with unique QR codes (qr_code, shipment_count).
pickup_requests	id, seller_id, assigned_staff_id	Pickup scheduling and collection lifecycle. Contains auto-generated tracking_number, status_history (JSON).
tracking_subscriptions	id	Customer alert subscribers. Contains tracking_number, email, phone, is_active.

12. Operational Runbook, Deployment Guards & Test Suite Verification

Scheduled Artisan Commands

COMMAND LINE	FREQUENCY	OPERATIONAL RESPONSIBILITY
php artisan tracking:sync-carriers	Every 15 minutes	Polls global carrier APIs for in-transit shipments and cascades delivery telemetry.
php artisan tracking:automate-domestic	Every 30 minutes	Audits domestic linehauls, identifies transit bottlenecks, and dispatches partner reminders.
php artisan queue:work --tries=3	Continuous daemon	Processes background SMS/email notifications, carrier webhooks, and PDF exports.

Automated Test Suite Verification (100% Passing)

The platform includes a rigorous test suite validating privacy safety, role authorization, MAWB cascades, HAWB printability, and multi-carrier webhooks:

Test Suite Execution Command: `php artisan test`
Verification Summary: 111 Tests Passed · 771 Assertions · 0 Failures · 100% Test Suite Health.

NETPACK LOGISTICS · DOCUMENTATION COMPLETE

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